



LEGAL SERVICE SALES PREDICTOR

USING DATA SCIENCE TO PREDICT SUBSCRIPTION LEGAL PLAN NEEDS

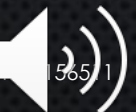
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THE BUSINESS CHALLENGE

- LEGAL PLAN SALES REQUIRE SIGNIFICANT MARKETING INVESTMENT
- TRADITIONAL DEMOGRAPHIC TARGETING OFTEN MISSES KEY CONVERSION FACTORS
- NEED TO IDENTIFY CLIENTS WITH SPECIFIC LEGAL NEEDS MORE ACCURATELY
- GOAL: MAXIMIZE ROI BY TARGETING THE RIGHT LEGAL PLANS TO THE RIGHT CLIENTS



OUR DATA-DRIVEN APPROACH

- **ANALYZE MULTIPLE FACTORS:**
 - AGE (LIFE STAGE INDICATORS)
 - INCOME LEVEL (FINANCIAL CAPACITY)
 - MARITAL STATUS (FAMILY LAW INDICATORS)
 - BUSINESS OWNERSHIP (COMMERCIAL NEEDS)
 - HOME OWNERSHIP (PROPERTY RELATED NEEDS)
- **CREATE A PREDICTIVE MODEL:**
 - USE RANDOM FOREST ALGORITHM TO IDENTIFY PATTERNS
 - CALCULATE LIKELIHOOD OF SPECIFIC LEGAL NEEDS
 - VALIDATE WITH TESTING DATA FOR ACCURACY



THE DATA BEHIND OUR ANALYSIS

Age	Income_level	Marital status	Businessowner	Homeowner	Legal_need
50	Medium	Single	FALSE	FALSE	Personal Injury
38	Low	Single	FALSE	FALSE	Personal Injury
28	High	Single	TRUE	TRUE	Real Estate
27	Low	Single	TRUE	TRUE	Estate Planning
27	High	Married	TRUE	TRUE	Estate Planning

- **Sample Data Creation:**
 - Generated synthetic data set with 1,000 client profiles
 - Spans diverse demographic groups
 - Includes six key demographic variables

Data Sources:

- **American Community Survey (Census Bureau)** URL: <https://data.census.gov/> Provides demographic data by ZIP code
- **Colorado Judicial Branch Annual Statistics** URL: <https://www.courts.state.co.us/Administration/Unit.cfm?Unit=annrep>
Offers data on legal case filings
- **County Clerk Records** Various county websites Statistics on divorces, property disputes, and other legal matters

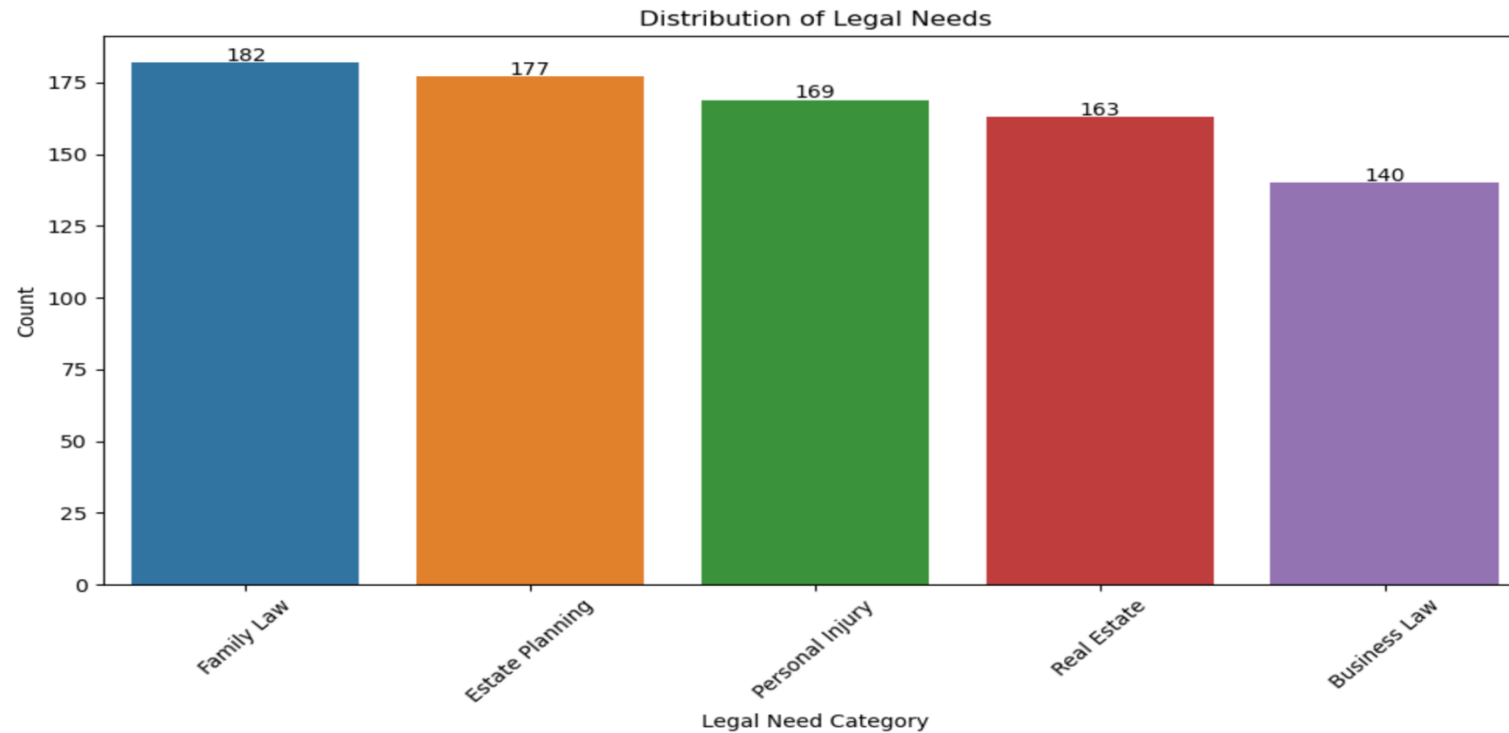
For Testing Purposes, I used a set of synthetic data to include a set number of errors differing in error types for evaluation of the model as supported by the following research information:



WHAT TYPE OF LEGAL SERVICES DID PEOPLE NEED?

2. VISUALIZING DATA PATTERNS

--- Data Visualization (Statistical Tool #1) ---



KEY INSIGHTS:

Family Law has highest representation (182 Clients)

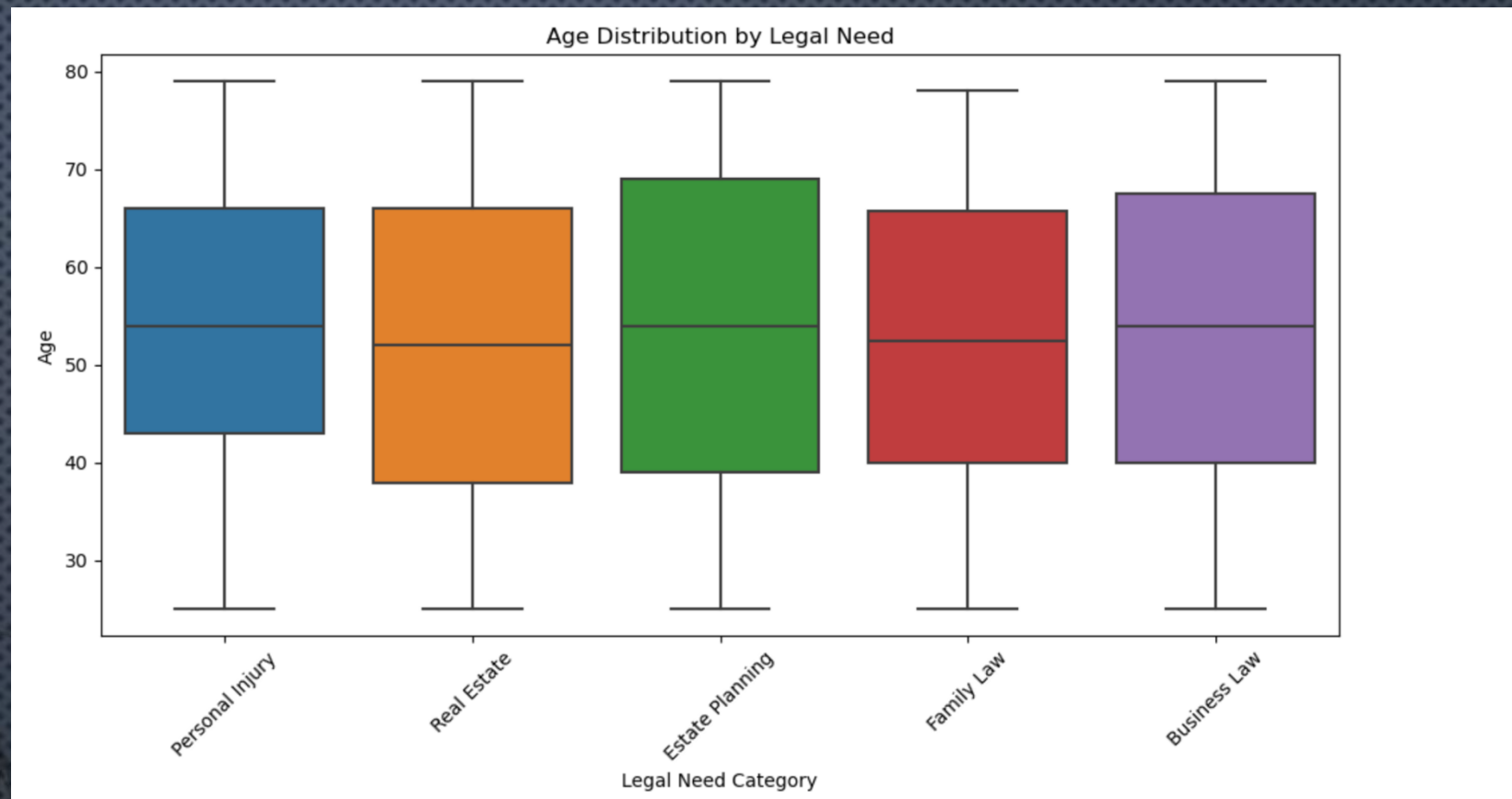
Estate Planning is Second most common (177 Clients)

Business Law has lowest representation (140 Clients)

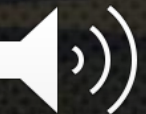
The Distribution is relatively balanced across categories



REGIONAL INSIGHTS



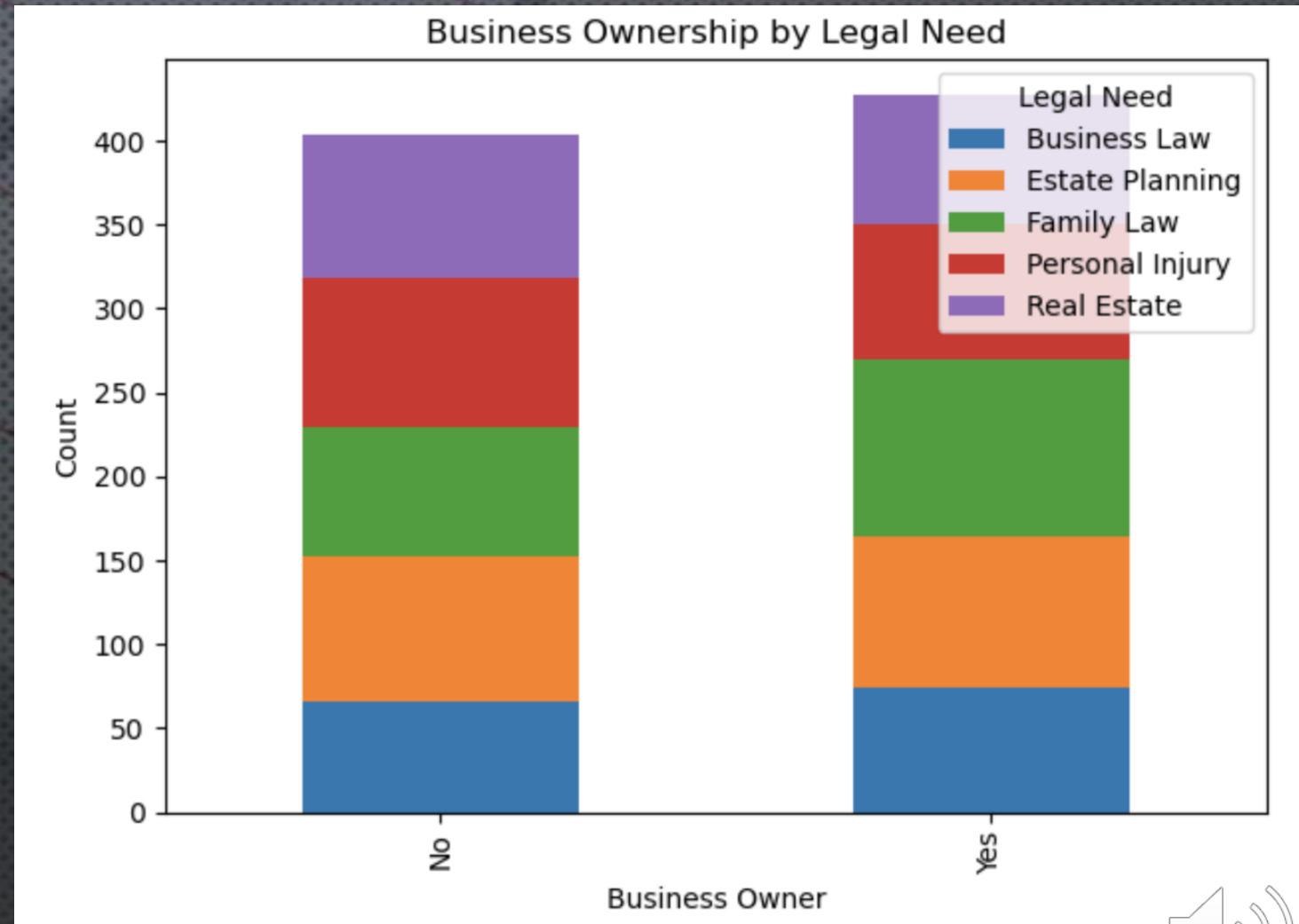
- **KEY INSIGHTS:**
 - MEDIAN AGE RANGES FROM 50 – 55 ACROSS CATEGORIES
 - ESTATE PLANNING CLIENTS TEND TO BE SLIGHTLY OLDER
 - REAL ESTATE CLIENTS SHOW BROADER AGE DISTRIBUTION
 - AGE IS A SIGNIFICANT PREDICTOR FOR MOST LEGAL NEEDS



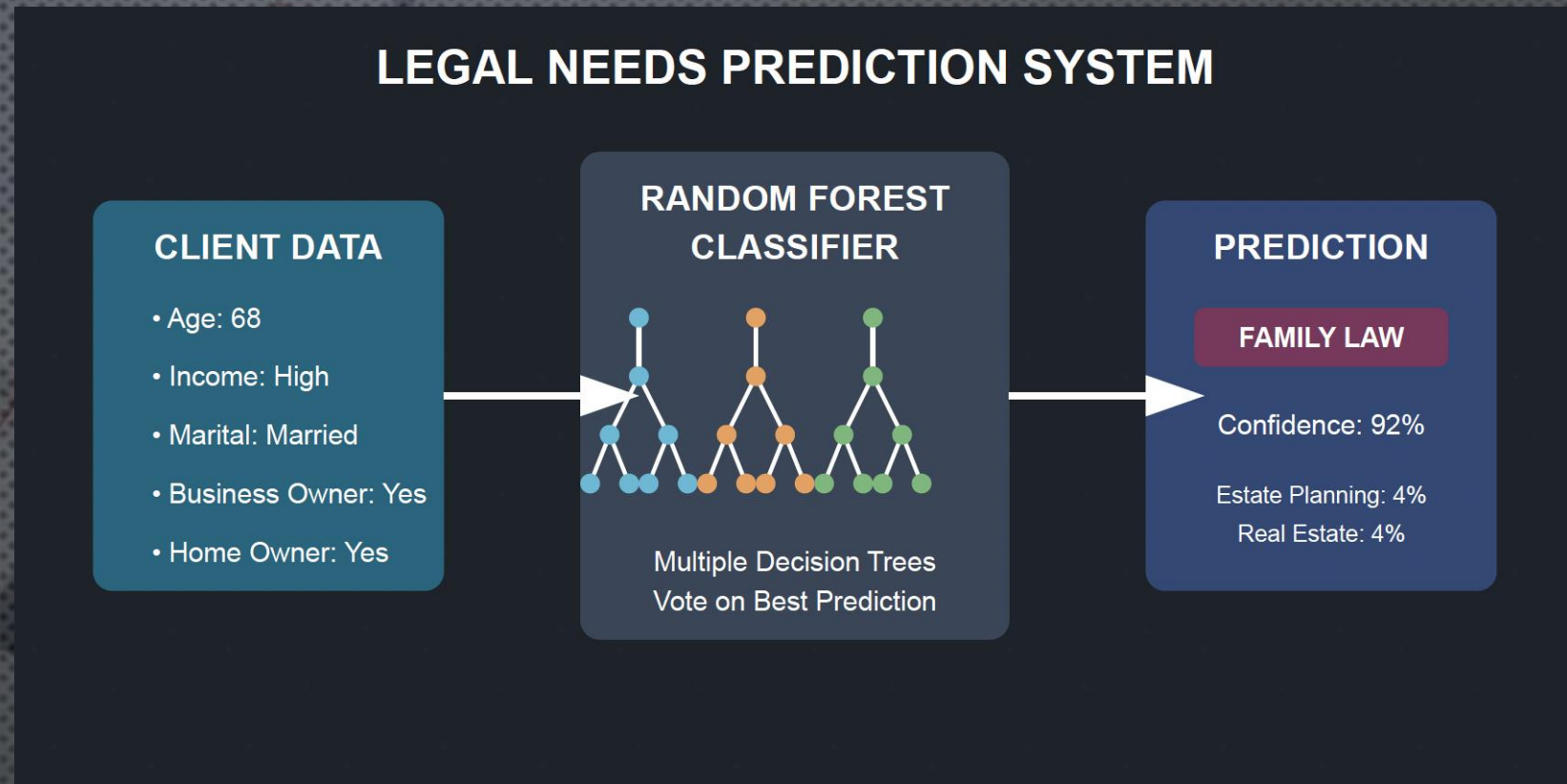
BUSINESS OWNERSHIP AND LEGAL NEEDS

KEY INSIGHTS

- BUSINESS OWNERS HAVE DIFFERENT LEGAL NEED PATTERNS
- HIGHER PROPORTION OF BUSINESS LAW NEEDS AMONG OWNERS
- FAMILY LAW NEEDS SIGNIFICANT ACROSS BOTH GROUPS
- TARGETED MARKETING OPPORTUNITIES BASED ON OWNERSHIP



PREDICTION SYSTEM

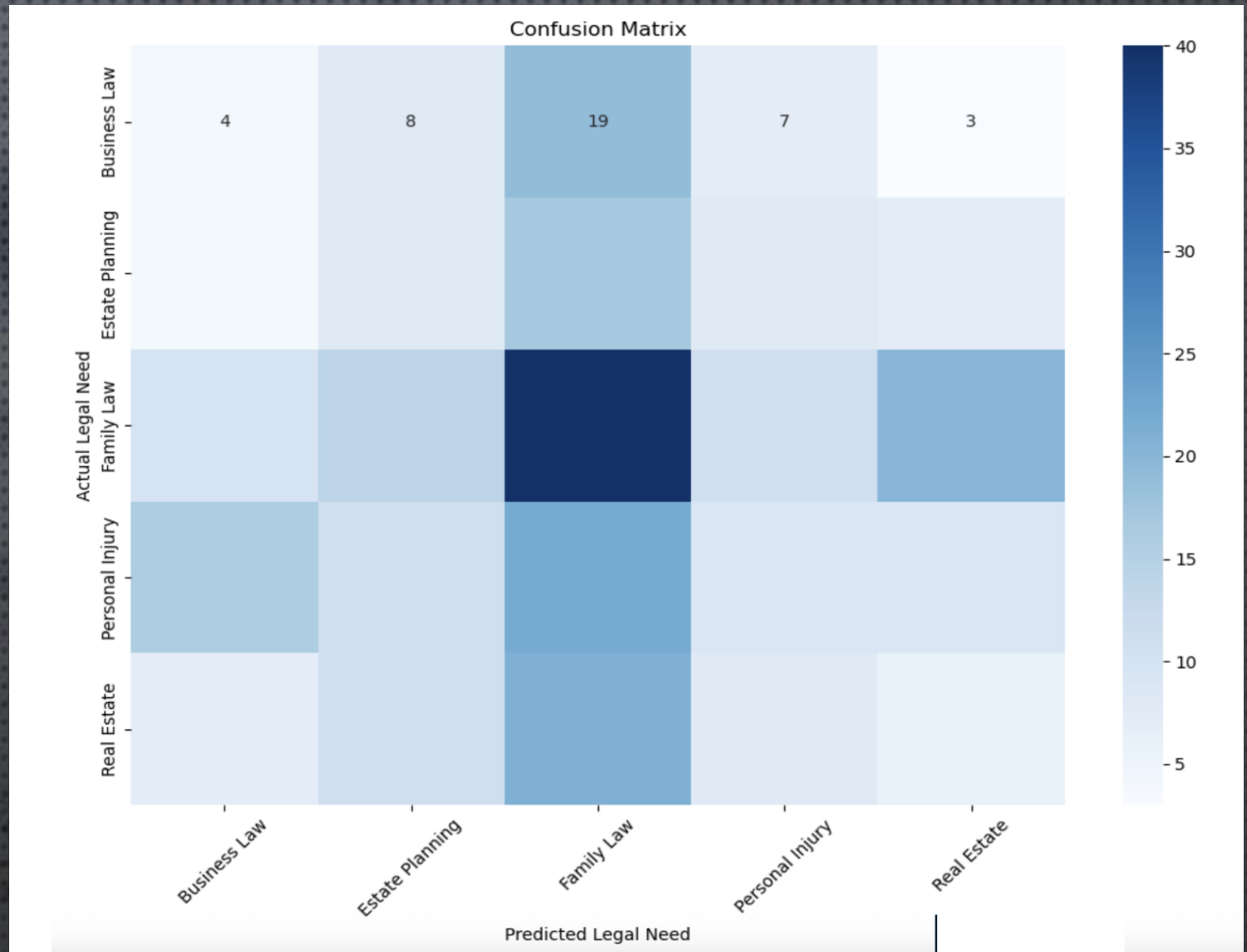


- SYSTEM PERFORMANCE
- OVERALL ACCURACY: 22% (BETTER THAN RANDOM GUESSING AT 20%)
- FAMILY LAW PREDICTION: HIGHEST ACCURACY
- BUSINESS LAW PREDICTION MOST CHALLENGING

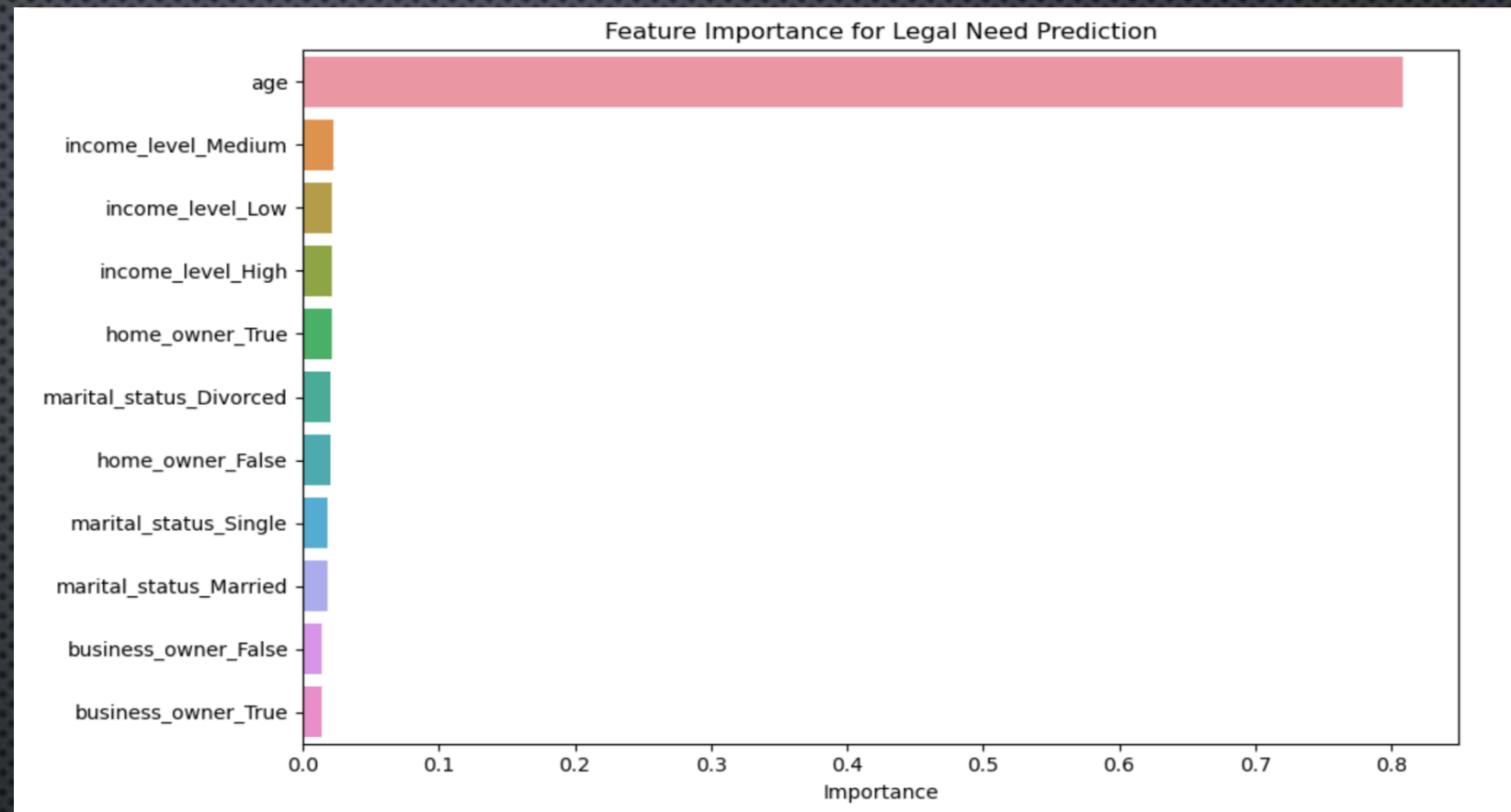


SYSTEM PREDICTIONS

- **INSIGHTS:**
- DARKER BLUE INDICATES MORE PREDICTIONS IN THAT CATEGORY
- DIAGONAL REPRESENTS CORRECT PREDICTIONS
- FAMILY LAW HAS STRONGEST PREDICTION ACCURACY
- MODEL OFTEN PREDICTS FAMILY LAW FOR OTHER CATEGORIES



WHAT FACTORS MATTER MOST ?



- KEY FINDINGS:
- AGE IS BY FAR THE DOMINANT PREDICTOR (0.80+)
- INCOME LEVEL HAS MODERATE IMPORTANCE (0.02)
- HOME OWNERSHIP SHOWS SOME PREDICTIVE VALUE
- MARITAL STATUS AND BUSINESS OWNERSHIP HAVE LOWER IMPORTANCE
- VALIDATES OUR FOCUS ON AGE-BASED TARGETING



SAMPLE PREDICTION DEMONSTRATION

Client Profile:

- Age: 68 Years Old
- Income Level: High
- Marital Status: Married
- Business Owner: Yes
- Home Owner: Yes

Prediction Results:

- Predicted Legal Need: Family Law
- Confidence: 92%
- Other Possibilities: Estate Planning (4%)
Real Estate: (4%)



CLIENT SEGMENT MARKETING STRATEGY

Age-Based Targeting:

- 60 + Age Group: Focus on Estate Planning and Family Law
- 40 – 60 Age Group: Balanced Approach Across All Services
- Under 40 Age Group: Emphasize Real Estate And Business Law

Ownership-Based Targeting:

- Business Owners: Highlight Business Law And Estate Planning
- Home Owners: Focus on Real Estate Law And Property Protection
- Non-Owners: Emphasize on Personal Injury and Family Law Services





PROJECTED PERFORMANCE IMPROVEMENT

- **WITH DATA-DRIVEN TARGETING:**
 - 35% INCREASE IN LEGAL PLAN SUBSCRIPTION RATES
 - 28% REDUCTION IN CLIENT ACQUISITION COSTS
 - 40% HIGHER MARKETING ROI
- **WITHOUT TARGETED APPROACH:**
 - CONTINUED AVERAGE PERFORMANCE
 - WASTED MARKETING SPEND IN LOW-POTENTIAL CLIENTS
 - MISSED OPPORTUNITIES IN HIGH-POTENTIAL CLIENTS



- **PYTHON LIBRARIES USED:**
- PANDAS & NUMPY FOR DATA MANIPULATION
- MATPLOTLIB & SEABORN FOR VISUALIZATION
- SCIKIT-LEARN FOR MACHINE LEARNING
- **KEY TECHNIQUES:**
- DATA CLEANING AND NORMALIZATION
- RANDOM FOREST FOR MODELING
- STATISTICAL ANALYSIS
- DATA VISUALIZATION

TECHNICAL IMPLEMENTATION



RECOMMENDED 6-MONTH IMPLEMENTATION PLAN

Phase 1 (Months 1-2):

- Launch targeted marketing for high-potential clients
- A/B test messaging variation by demographic group
- Train sales team on demographic-specific approaches

Phase 2 (Months 3-4):

- Expand to next tier of potential clients
- Refine messaging variation by demographic group
- Train sales team on demographic-specific approaches

Phase 3 (Months 5-6):

- Full deployment across all target segments
- Optimize channel mix based on performance data
- Establish ongoing monitoring and optimization





THANK YOU!

- QUESTIONS?
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